Debrief

Run by

Stage 13 dry run (rehearsal, chromium)

Scenario	Malignant hyperthermia
Patient	M 35 y 80 kg
Started	10/11/2026, 5:13:53 PM
Duration	40:06 simulated · 10 min 03 s on the clock

Objectives

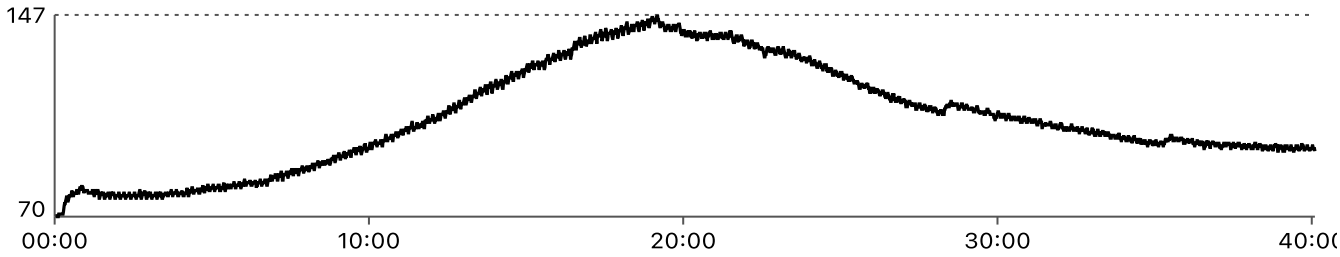
1. **✕ Not met** Call the MH crisis (stop sevoflurane, O_2 100 % at 10 L/min, hyperventilate) within 120 s of the early signs ($EtCO_2$ above 45 mmHg at the same ventilation)
   The state "Trigger off: sevoflurane stopped, O_2 100 % at 10 L/min, ventilation 20 × 600" 12 min 57 s after the state "Early signs: $EtCO_2$ rising at the same ventilation, tachycardia" (06:52 → 19:49; within 2 min 00 s asked).
2. **✓ Met** Give dantrolene 2.5 mg/kg within 300 s of calling the crisis
   Dantrolene 1 min 02 s after the state "Trigger off: sevoflurane stopped, O_2 100 % at 10 L/min, ventilation 20 × 600" (19:49 → 20:51; within 5 min 00 s asked).
3. **– Not assessable** Repeat dantrolene 2.5 mg/kg within 600 s of the first dose while $EtCO_2$ is still above 45 mmHg
   This objective has no rule: the instructor judges it.
4. **✓ Met** Treat the potassium (calcium chloride, insulin with dextrose) within 300 s of K^+ rising above 5.5 mmol/L
   Calcium chloride 10 % or Insulin + dextrose 4 min 38 s after K^+ above 5.5 mmol/L (17:12 → 21:49; within 5 min 00 s asked).
5. **✓ Met** Bring the ventilation back to 12 × 500 within 300 s of $EtCO_2$ falling below 40 mmHg
   The state "Ventilation back to 12 × 500, O_2 100 %" 2 min 05 s after $EtCO_2$ below 40 mmHg after the state "Trigger off: sevoflurane stopped, O_2 100 % at 10 L/min, ventilation 20 × 600" (37:01 → 39:05; within 5 min 00 s asked).

Measured facts

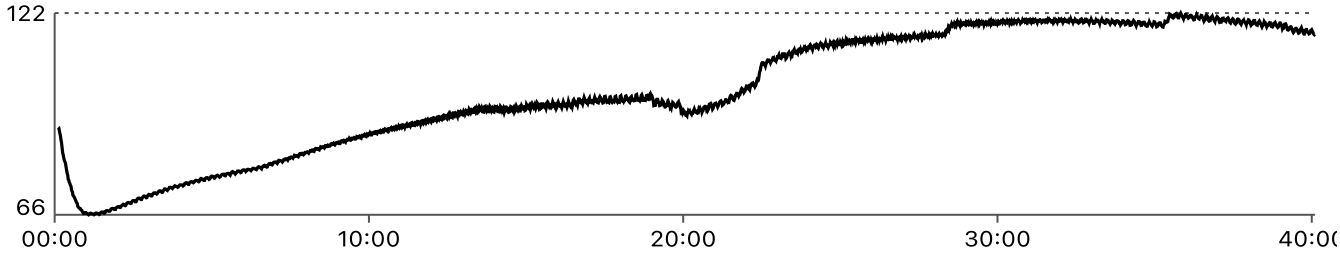
Lowest SpO_2	96 % at 00:10
Lowest ART M / NIBP M	66 mmHg at 01:15
Highest $EtCO_2$	81 mmHg at 18:36
Highest T1 / T2	39.5 °C at 28:30
Labs (the model): last (lowest–highest)	K^+ 4.5 mmol/L (4.2–5.9) · Lactate 6.4 mmol/L (1.0–7.4) · pH 7.34 (7.09–7.41) · Glucose 5.9 mmol/L (5.6–17.2) · Tcore 39.4 °C (36.7–39.5)

Trends

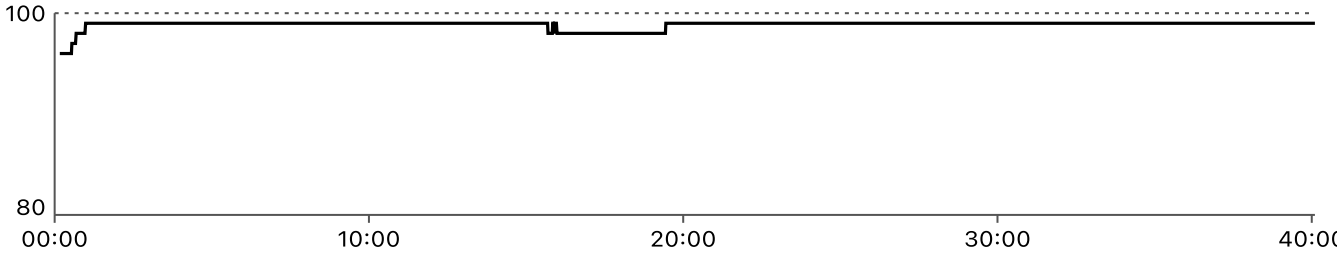
HR bpm



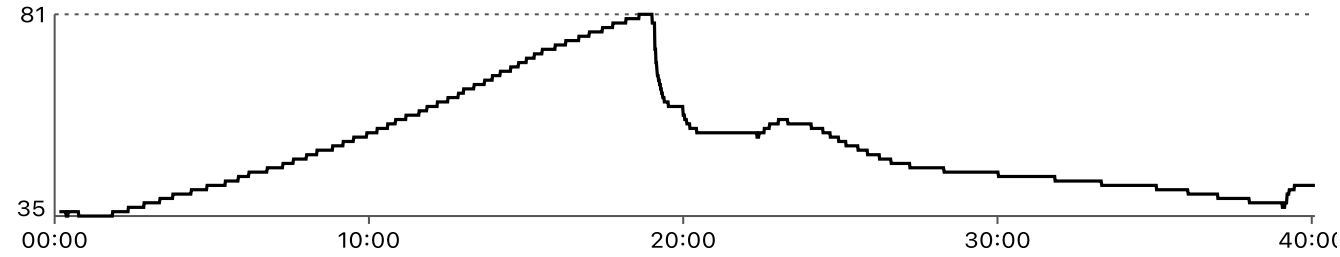
ART M / NIBP M mmHg



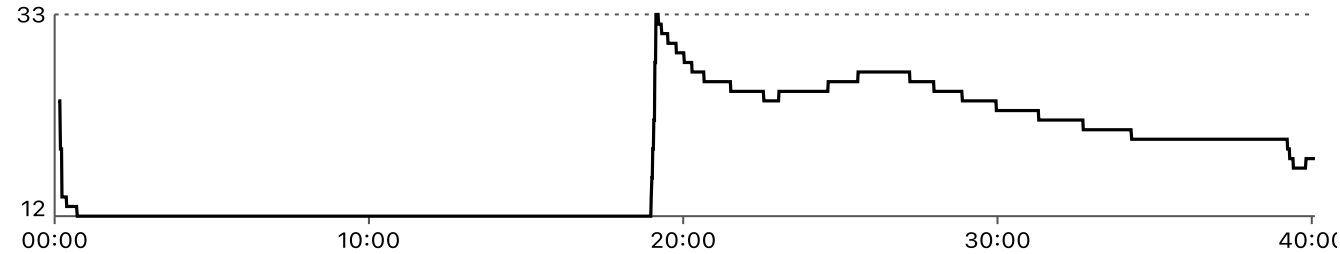
SpO_2 %

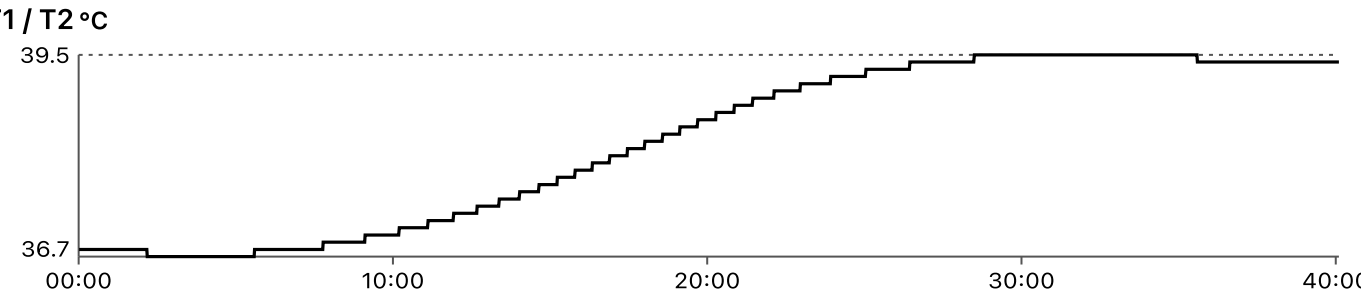


$EtCO_2$ mmHg



awRR / RR /min





Drugs and fluids

Time	Given
00:00	Propofol 2 mg/kg IV
00:00	Succinylcholine 1.5 mg/kg IV
04:05	Rocuronium 0.6 mg/kg IV
20:51	Dantrolene 2.5 mg/kg IV
21:19	Crystalloid 2000 mL over 15 min
21:49	Calcium chloride 10 % 10 mg/kg IV
21:49	Insulin + dextrose 10 units IV
22:21	Sodium bicarbonate 8.4 % 1 mmol/kg IV
29:50	Dantrolene 2.5 mg/kg IV

- Propofol: 160 mg in 1 bolus
- Succinylcholine: 120 mg in 1 bolus
- Rocuronium: 48 mg in 1 bolus
- Dantrolene: 400 mg in 2 boluses
- Calcium chloride 10 %: 800 mg in 1 bolus
- Insulin + dextrose: 10 units in 1 bolus
- Sodium bicarbonate 8.4 %: 1 mmol/kg in 1 bolus

Timeline

Time	Kind	Event
00:00	Scenario	Scenario loaded: Malignant hyperthermia
00:00	Scenario	Scenario state: Induced: propofol, succinylcholine 1.5 mg/kg (masseter rigidity), intubated; sevoflurane 2 %
00:00	Scenario	Propofol 2 mg/kg IV
00:00	Scenario	Succinylcholine 1.5 mg/kg IV
00:00	Scenario	Tracheal tube in place
00:00	Scenario	Ventilator: RR 12 /min, VT 500 mL, PEEP 5 cmH_2O, FiO_2 50 %
00:00	Scenario	Sevoflurane 2 % at 2 L/min fresh gas
00:00	Scenario	Malignant hyperthermia: severe
00:01	Action	Speed ×4
00:00	Patient	Ventilated breaths began
04:05	Scenario	Scenario state: Surgery under way: sevoflurane 2 %, rocuronium
04:05	Scenario	Rocuronium 0.6 mg/kg IV
06:52	Scenario	Scenario state: Early signs: $EtCO_2$ rising at the same ventilation, tachycardia
18:46	Scenario	Scenario state: Established MH: hyperthermia, acidosis, rising potassium
19:49	Scenario	Scenario state: Trigger off: sevoflurane stopped, O_2 100 % at 10 L/min, ventilation 20 × 600
19:49	Scenario	Sevoflurane vaporiser off
19:49	Scenario	Ventilator: RR 20 /min, VT 600 mL, PEEP 5 cmH_2O, FiO_2 100 %
20:51	Scenario	Scenario state: Dantrolene 2.5 mg/kg given
20:51	Scenario	Dantrolene 2.5 mg/kg IV
20:53	Scenario	Scenario state: MH treatment under way
21:19	Scenario	Scenario state: Cooling: 2 L saline at room temperature running
21:19	Scenario	Crystalloid 2000 mL over 15 min
21:22	Scenario	Scenario state: MH treatment under way
21:49	Scenario	Scenario state: Calcium chloride 10 mg/kg and insulin 10 U with dextrose given
21:49	Scenario	Calcium chloride 10 % 10 mg/kg IV
21:49	Scenario	Insulin + dextrose 10 units IV
21:51	Scenario	Scenario state: MH treatment under way
22:21	Scenario	Scenario state: Sodium bicarbonate 1 mmol/kg given
22:21	Scenario	Sodium bicarbonate 8.4 % 1 mmol/kg IV
22:23	Scenario	Scenario state: MH treatment under way
29:50	Scenario	Scenario state: Dantrolene 2.5 mg/kg given
29:50	Scenario	Dantrolene 2.5 mg/kg IV
29:53	Scenario	Scenario state: MH treatment under way
30:53	Scenario	Scenario state: Recovering: $EtCO_2$ below 45
39:05	Scenario	Scenario state: Ventilation back to 12 × 500, O_2 100 %
39:05	Scenario	Ventilator: RR 12 /min, VT 500 mL, PEEP 5 cmH_2O, FiO_2 100 %

Bookmarks

No bookmarks.

Notes

Appendix: one row per minute

Time	HR bpm	ART M / NIBP M mmHg	SpO_2 %	$EtCO_2$ mmHg	awRR / RR /min	T1 / T2 °C	K^+ mmol/L	Lactate mmol/L	pH	Glucose mmol/L	Tcore °C
00:00	71	90	96	36	24	36.8	4.2	1.0	7.41	5.6	36.8
01:00	80	67	99	35	12	36.8	4.3	1.0	7.40	5.6	36.8
02:00	78	68	99	36	12	36.8	4.5	1.0	7.39	5.6	36.7
03:00	78	72	99	38	12	36.7	4.6	1.0	7.38	5.6	36.7
04:00	79	74	99	40	12	36.7	4.7	1.0	7.37	5.6	36.7
05:00	81	77	99	42	12	36.7	4.7	1.1	7.35	5.6	36.8
06:00	82	79	99	44	12	36.8	4.7	1.2	7.34	5.6	36.8
07:00	85	81	99	46	12	36.8	4.6	1.3	7.32	5.6	36.8
08:00	88	83	99	48	12	36.9	4.7	1.4	7.30	5.6	36.9
09:00	93	86	99	51	12	36.9	4.7	1.6	7.28	5.7	37.0
10:00	96	88	99	54	12	37.0	4.8	1.8	7.26	5.7	37.1
11:00	102	90	99	57	12	37.1	4.8	2.1	7.24	5.8	37.2
12:00	107	92	99	60	12	37.3	4.9	2.4	7.22	5.8	37.4
13:00	113	94	99	63	12	37.4	5.0	2.7	7.19	5.9	37.5
14:00	120	95	99	67	12	37.5	5.1	3.0	7.17	6.0	37.7
15:00	126	96	99	70	12	37.7	5.3	3.3	7.15	6.0	37.9
16:00	131	97	98	74	12	37.9	5.4	3.7	7.13	6.1	38.0
17:00	136	98	98	76	12	38.1	5.5	4.0	7.12	6.2	38.2
18:00	141	99	98	79	12	38.2	5.6	4.3	7.10	6.3	38.4
19:00	146	99	98	81	16	38.4	5.8	4.7	7.14	6.4	38.6
20:00	141	95	99	58	29	38.6	5.8	5.0	7.21	6.5	38.7
21:00	139	97	99	54	26	38.8	5.9	5.3	7.22	6.6	38.9
22:00	137	101	99	54	25	38.9	5.9	5.7	7.22	17.2	39.0
23:00	134	110	99	56	24	39.1	5.8	6.0	7.27	16.9	39.1
24:00	130	112	99	56	25	39.2	5.8	6.2	7.27	16.3	39.2
25:00	125	114	99	52	26	39.2	5.7	6.5	7.29	15.7	39.3
26:00	119	114	99	49	27	39.3	5.6	6.7	7.31	15.0	39.4
27:00	115	115	99	47	27	39.4	5.5	6.9	7.32	14.2	39.4
28:00	112	116	99	46	25	39.4	5.4	7.1	7.32	13.4	39.5
29:00	112	119	99	45	24	39.5	5.3	7.2	7.32	12.5	39.5
30:00	109	119	99	45	23	39.5	5.2	7.3	7.32	11.7	39.5
31:00	108	120	99	44	23	39.5	5.2	7.4	7.32	11.0	39.5
32:00	105	120	99	43	22	39.5	5.1	7.4	7.32	10.2	39.5
33:00	103	120	99	43	21	39.5	5.0	7.4	7.33	9.5	39.5
34:00	100	120	99	42	21	39.5	4.9	7.3	7.33	8.9	39.5
35:00	99	119	99	42	20	39.5	4.8	7.2	7.34	8.3	39.4
36:00	99	121	99	41	20	39.4	4.8	7.0	7.34	7.7	39.4
37:00	98	121	99	40	20	39.4	4.7	6.9	7.35	7.2	39.4
38:00	98	120	99	39	20	39.4	4.6	6.7	7.36	6.7	39.4

Time	HR bpm	ART M / NIBP M mmHg	SpO_2 %	$EtCO_2$ mmHg	awRR / RR /min	T1 / T2 °C	K^+ mmol/L	Lactate mmol/L	pH	Glucose mmol/L	Tcore °C
39:00	97	119	99	38	20	39.4	4.6	6.6	7.37	6.3	39.4
40:00	96	117	99	42	18	39.4	4.5	6.4	7.33	5.9	39.4